Supplementary materials for the paper "Factor Modeling for High-Dimensional Time Series: Inference for the Number of Factors"- Proof of Theorem 3 and 4

First, we need to show the rates of λ_j and λ_j^* for $\mathbf{M}$ and $\mathbf{M}^*$ respectively, as in (6.23) and (6.24). We state the Wielandt's inequality below.

Lemma S.1 *Let* $\mathbf{A} \in \mathbb{R}^{r\times r}$ *be a symmetric matrix such that*

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{C}' & \mathbf{D} \end{pmatrix}, \quad \textit{with } \mathbf{B} \in \mathbb{R}^{r_1\times r_1},\ \mathbf{D} \in \mathbb{R}^{r_2\times r_2} \quad \textit{and} \quad \lambda_{r_1}(\mathbf{B}) > \lambda_1(\mathbf{D}).$$

Then for $1 \le j \le r_2$*,*

$$0 \le \lambda_j(\mathbf{D}) - \lambda_{r_1+j}(\mathbf{A}) \le \frac{\lambda_1(\mathbf{C}\mathbf{C}')}{\lambda_{r_1}(\mathbf{B}) - \lambda_j(\mathbf{D})}.$$

Hereafter we suppress k_0 and write $\mathbf{W}_y = \mathbf{W}_y(k_0)$, $\mathbf{W}_i = \mathbf{W}_i(k_0)$, $\mathbf{W}_{21} = \mathbf{W}_{21}(k_0)$ and $\mathbf{W}_{12} = \mathbf{W}_{12}(k_0)$. Similarly $\mathbf{W}_{\mathbf{y}^*} = \mathbf{W}_{\mathbf{y}^*}(k_0)$. Refer to the proof of Theorem 2 and the statement of Theorem 3 for definitions.

Proof of (6.23) and (6.24). With the notations in the proof of Theorem 2 and section 4, we have $\mathbf{M} = \mathbf{W}_y\mathbf{W}_y'$, where we can write, by assumption C6',

$$\begin{aligned} \mathbf{W}_y &= \mathbf{A}_1\mathbf{L}_1 + \mathbf{A}_2\mathbf{L}_2, \quad \text{with} \\ \mathbf{L}_1 &= \mathbf{W}_1(\mathbf{I}_{k_0} \otimes \mathbf{A}_1') + \mathbf{W}_{12}(\mathbf{I}_{k_0} \otimes \mathbf{A}_2'), \\ \mathbf{L}_2 &= \mathbf{W}_2(\mathbf{I}_{k_0} \otimes \mathbf{A}_2') + \mathbf{W}_{21}(\mathbf{I}_{k_0} \otimes \mathbf{A}_1'). \end{aligned}$$

To find the order of λ_j for $j = 1, \cdots, r_1$, using a standard singular value inequality, for $j = 1, \cdots, r_1$,

$$\lambda_j = \lambda_j(\mathbf{M}) = \sigma_j^2(\mathbf{W}_y) \ge \{\sigma_j(\mathbf{A}_1\mathbf{L}_1) - \sigma_1(\mathbf{A}_2\mathbf{L}_2)\}^2 = \{\sigma_j(\mathbf{L}_1) - \sigma_1(\mathbf{L}_2)\}^2,$$

where assumptions C5' ensures that $\mathbf{L}_1$ has the j-th singular value dominates those of $\mathbf{L}_2$, since $\delta_2 > \delta_1 = 0$. Hence, using the same singular value inequality,

$$\begin{aligned} \lambda_j \ge \{\sigma_j(\mathbf{L}_1) - \sigma_1(\mathbf{L}_2)\}^2 &\asymp \sigma_j^2(\mathbf{L}_1) \\ &\ge \{\sigma_j(\mathbf{W}_1(\mathbf{I}_{k_0} \otimes \mathbf{A}_1')) - \sigma_1(\mathbf{W}_{12}(\mathbf{I}_{k_0} \otimes \mathbf{A}_2'))\}^2 \\ &\asymp \sigma_j^2(\mathbf{W}_1(\mathbf{I}_{k_0} \otimes \mathbf{A}_1')) = \sigma_j^2(\mathbf{W}_1) \\ &= \lambda_j\left(\sum_{k=1}^{k_0} \boldsymbol{\Sigma}_1(k)\boldsymbol{\Sigma}_1(k)'\right) \\ &\ge \lambda_j(\boldsymbol{\Sigma}_1(1)\boldsymbol{\Sigma}_1(1)') \ge \|\boldsymbol{\Sigma}_1(1)\|_{\min} \asymp p^2, \end{aligned}$$

where the start of the second line follows from condition C5' and the fact that $\|\mathbf{W}_{12}\|, \|\mathbf{W}_{21}\| = O(p^{1-\delta_2/2}) = o(\|\mathbf{W}_1\|)$. This shows that $\lambda_j \asymp p^2$ for $j = 1, \cdots, r_1$, which is the largest possible order.

Also, $\mathbf{M} = \mathbf{W}_y \mathbf{W}_y' = \mathbf{A}\mathbf{L}\mathbf{A}'$, where

$$\mathbf{A} = (\mathbf{A}_1 \ \mathbf{A}_2), \quad \mathbf{L} = \begin{pmatrix} \mathbf{L}_1\mathbf{L}_1' & \mathbf{L}_1\mathbf{L}_2' \\ \mathbf{L}_2\mathbf{L}_1' & \mathbf{L}_2\mathbf{L}_2' \end{pmatrix}, \tag{S.1}$$

and $\mathbf{M}$ and $\mathbf{L}$ shares the same eigenvalues, as $\mathbf{A}'\mathbf{A} = \mathbf{I}$. Hence $\lambda_j = \lambda_j(\mathbf{M}) = \lambda_j(\mathbf{L})$. By Lemma S.1 applying on $\mathbf{L}$, we have for $j = 1, \cdots, r_2$,

$$\lambda_{r_1+j}(\mathbf{L}) \le \lambda_j(\mathbf{L}_2\mathbf{L}_2') = \sigma_j^2(\mathbf{L}_2), \tag{S.2}$$

$$\lambda_{r_1+j}(\mathbf{L}) \ge \sigma_j^2(\mathbf{L}_2) - \frac{\sigma_1^2(\mathbf{L}_1\mathbf{L}_2')}{\sigma_{r_1}^2(\mathbf{L}_1) - \sigma_1^2(\mathbf{L}_2)}. \tag{S.3}$$

From the definition of $\mathbf{L}_1$ and $\mathbf{L}_2$, we have $\mathbf{L}_1\mathbf{L}_2' = \mathbf{W}_1\mathbf{W}_{21}' + \mathbf{W}_2\mathbf{W}_{12}'$. By condition C5' and the Cauchy-Schwarz inequality for spectral norm, we have

$$\|\mathbf{W}_2\mathbf{W}_{12}'\| \le \|\mathbf{W}_2\|\|\mathbf{W}_{12}\| = O(p^{1-\delta_2} \cdot p^{1-\delta_2/2}) = O(p^{2-3\delta_2/2}) = o(\|\mathbf{L}_1\|\|\mathbf{L}_2\|),$$

since $\|\mathbf{L}_1\| = \sigma_1(\mathbf{L}_1) \asymp \sigma_1(\mathbf{W}_1) \asymp p$, and $\|\mathbf{L}_2\| = \sigma_1(\mathbf{L}_2) \asymp \max(\|\mathbf{W}_2\|, \|\mathbf{W}_{21}\|)$ with $\|\mathbf{W}_2\| \asymp p^{1-\delta_2}$. Hence asymptotically we can always find a constant $C_2 > 0$ such that

$$\|\mathbf{W}_2\mathbf{W}_{12}'\| \le C_2\|\mathbf{L}_1\|\|\mathbf{L}_2\|. \tag{S.4}$$

Case 1. If $\|\mathbf{W}_{21}\|_{\min} \asymp p^{1-c\delta_2}$ ($\asymp \|\mathbf{W}_{21}\|$ by condition C5') for $1/2 \le c < 1$, then $\|\mathbf{W}_2\| \asymp p^{1-\delta_2} = o(\|\mathbf{W}_{21}\|)$. Hence $\|\mathbf{L}_2\| \asymp \|\mathbf{W}_{21}\| \asymp p^{1-c\delta_2}$. The assumption $\|\mathbf{W}_1\mathbf{W}_{21}'\| \le q\|\mathbf{W}_1\|_{\min}\|\mathbf{W}_{21}\|$ for $0 < q < 1$ then ensures that

$$\begin{aligned} \|\mathbf{W}_1\mathbf{W}_{21}'\| &\le q\|\mathbf{W}_1\|_{\min}\|\mathbf{W}_{21}\| = (q\sigma_{r_1}(\mathbf{W}_1)/\sigma_1(\mathbf{W}_1))\|\mathbf{W}_1\|\|\mathbf{W}_{21}\| \\ &\le (q\sigma_{r_1}(\mathbf{W}_1)/\sigma_1(\mathbf{W}_1))(\|\mathbf{L}_1\| + \|\mathbf{W}_{12}\|)(\|\mathbf{L}_2\| + \|\mathbf{W}_2\|) \\ &= (q\sigma_{r_1}(\mathbf{W}_1)/\sigma_1(\mathbf{W}_1))\|\mathbf{L}_1\|\|\mathbf{L}_2\|(1 + o(1))(1 + o(1)) \\ &\le C_1\|\mathbf{L}_1\|\|\mathbf{L}_2\|, \end{aligned} \tag{S.5}$$

where $0 < C_1 < \sigma_{r_1}(\mathbf{L}_1)/\sigma_1(\mathbf{L}_1)$, when n, p are sufficiently large. Combining (S.4) and (S.5), we then have

$$\begin{aligned} \sigma_1(\mathbf{L}_1\mathbf{L}_2') &\le \|\mathbf{W}_1\mathbf{W}_{21}'\| + \|\mathbf{W}_2\mathbf{W}_{12}'\| \le (C_1 + C_2)\sigma_1(\mathbf{L}_1)\sigma_1(\mathbf{L}_2) \\ &= C\sigma_1(\mathbf{L}_1)\sigma_1(\mathbf{L}_2), \end{aligned} \tag{S.6}$$

where C_2 in (S.4) is chosen so that $0 < C = C_1 + C_2 < \sigma_{r_1}(\mathbf{L}_1)/\sigma_1(\mathbf{L}_1)$.

Case 2. If $\|\mathbf{W}_{21}\|_{\min} = o(p^{1-\delta_2})$, then $\|\mathbf{W}_{21}\| = o(\|\mathbf{W}_2\|)$. Hence $\|\mathbf{L}_2\| \asymp \|\mathbf{W}_2\| \asymp p^{1-\delta_2}$. Hence

$$\|\mathbf{W}_1\mathbf{W}_{21}'\| \leq \|\mathbf{W}_1\|\|\mathbf{W}_{21}\| = o(\|\mathbf{L}_1\|\|\mathbf{L}_2\|),$$

and thus asymptotically we can always find a constant $C_1 > 0$ such that

$$\|\mathbf{W}_1\mathbf{W}_{21}'\| \leq C_1\|\mathbf{L}_1\|\|\mathbf{L}_2\|. \tag{S.7}$$

Combining (S.4) and (S.7), and choosing C_1 and C_2 to be small enough, (S.6) will hold.

With (S.6), (S.3) becomes

$$\begin{aligned}\lambda_{r_1+j}(\mathbf{L}) &\geq \sigma_j^2(\mathbf{L}_2) - \frac{C^2\sigma_1^2(\mathbf{L}_1)\sigma_1^2(\mathbf{L}_2)}{\sigma_{r_1}^2(\mathbf{L}_1) - \sigma_1^2(\mathbf{L}_2)} \\ &= \sigma_j^2(\mathbf{L}_2)\left(1 - \frac{C^2}{\sigma_{r_1}^2(\mathbf{L}_1)/\sigma_1^2(\mathbf{L}_1) - o(1)}\right) \geq C'\sigma_j^2(\mathbf{L}_2)\end{aligned}$$

for some constant $C' > 0$. Together with (S.2), we have $\lambda_{r_1+j}(\mathbf{L}) \asymp \sigma_j^2(\mathbf{L}_2)$ for $j = 1, \cdots, r_2$.

Hence, if $\|\mathbf{W}_{21}\|_{\min} = o(p^{1-\delta_2})$, then $\mathbf{W}_2$ dominates and from Case 2 above,

$$\lambda_{r_1+j}(\mathbf{L}) \asymp \sigma_j^2(\mathbf{L}_2) \asymp \sigma_j^2(\mathbf{W}_2) \asymp p^{2-2\delta_2}.$$

If $\|\mathbf{W}_{21}\|_{\min} \asymp p^{1-c\delta_2}$ for $1/2 \leq c < 1$, then $\mathbf{W}_{21}$ dominates, and from Case 1 above,

$$\lambda_{r_1+j}(\mathbf{L}) \asymp \sigma_j^2(\mathbf{L}_2) \asymp \sigma_j^2(\mathbf{W}_{21}) \asymp p^{2-2c\delta_2},$$

which completes the proof of (6.23).

For λ_j^* with $j = 1, \cdots, r_2$, note that $\mathbf{M}^* = \mathbf{W}_{y^*}\mathbf{W}_{y^*}'$, where $\mathbf{W}_{y^*}$ is formed similar to $\mathbf{W}_y$, but by $\mathbf{y}_t^{0*} = \mathbf{y}_t - \mathbf{A}_1\mathbf{A}_1'\mathbf{y}_t$ (similar to (4.14), but $\widehat{\mathbf{A}}_1$ is replaced by $\mathbf{A}_1$). With condition C6', we can show by simple algebra that

$$\mathbf{W}_{y^*} = \mathbf{A}_2\mathbf{W}_2(\mathbf{I}_{k_0} \otimes \mathbf{A}_2'). \tag{S.8}$$

Hence,

$$\lambda_j^* = \sigma_j^2(\mathbf{W}_{y^*}) = \sigma_j^2(\mathbf{W}_2) \asymp p^{2-2\delta_2},$$

which completes the proof (6.24). $\square$

Next, we state another lemma, which is a slightly modified version of Theorem 4.1 of Stewart (1973).

Lemma S.2 *Let $\mathbf{Y}$ be a real symmetric $p \times p$ matrix, and let $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ be orthogonal with $\mathbf{X}_1 \in \mathbb{R}^{p \times l}$. Partition $\mathbf{X}'\mathbf{Y}\mathbf{X}$ conformally with $\mathbf{X}$ in the form*

$$\mathbf{X}'\mathbf{Y}\mathbf{X} = \begin{pmatrix} \mathbf{Y}_{11} & \mathbf{Y}_{12} \\ \mathbf{Y}_{12}' & \mathbf{Y}_{22} \end{pmatrix}.$$

Define the operator $T : \mathbb{R}^{(p-l)\times l} \to \mathbb{R}^{(p-l)\times l}$ by

$$T\mathbf{P} = \mathbf{P}\mathbf{Y}_{11} - \mathbf{Y}_{22}\mathbf{P},$$

and suppose T is invertible with $\theta = \|T^{-1}\|^{-1}$. Define also $\eta = \|\mathbf{Y}_{12}\|$. If

$$\eta < \theta/2, \tag{S.9}$$

there is a matrix $\mathbf{P} \in \mathbb{R}^{(p-l)\times l}$ with

$$\|\mathbf{P}\| \leq 2\eta/\theta$$

such that the columns of

$$\widetilde{\mathbf{X}}_1 = (\mathbf{X}_1 + \mathbf{X}_2\mathbf{P})(\mathbf{I} + \mathbf{P}'\mathbf{P})^{-1/2}$$

span an invariant subspace of $\mathbf{Y}$.

For our purpose, l above is either r_1 or r_2, so that l is finite. Define the relation

$$\operatorname{sep}(\mathbf{A}, \mathbf{C}) = \min_{\lambda \in \lambda(\mathbf{A}), \mu \in \lambda(\mathbf{C})} |\lambda - \mu|.$$

Gathering the inequalities involving the operator T in Stewart (1973), since $\mathbf{Y}_{11}$ and $\mathbf{Y}_{22}$ are symmetric, we have

$$\operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22})/l \leq \theta \leq \operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}),$$

implying that

$$\theta \asymp \operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}).$$

Hence, if we can show that

$$\|\mathbf{Y}_{12}\| = \|\mathbf{X}_1'\mathbf{Y}\mathbf{X}_2\| = o_P(\operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22})), \tag{S.10}$$

then (S.9) is satisfied asymptotically, so that Lemma S.2 says that there exists $\mathbf{P}$ with $\|\mathbf{P}\| = O_P(\|\mathbf{Y}_{12}\|/\operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}))$ such that the columns of $\widehat{\mathbf{X}}_1 = (\mathbf{X}_1 + \mathbf{X}_2\mathbf{P})(\mathbf{I} + \mathbf{P}'\mathbf{P})^{-1/2}$ span an invariant subspace of $\mathbf{Y}$, with

$$\begin{aligned} \|\widehat{\mathbf{X}}_1 - \mathbf{X}_1\| &= \|\mathbf{X}_2\mathbf{P}\| + o_P(\|\mathbf{P}\|) \\ &= O_P(\|\mathbf{P}\|) = O_P(\|\mathbf{Y}_{12}\|/\operatorname{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22})). \end{aligned} \tag{S.11}$$

The proof of Theorem 3 involves suitable substitution of $\mathbf{Y}$, $\mathbf{X}_1$, $\mathbf{X}_2$, and the application of (S.11), after proving that (S.10) is satisfied.

Proof of Theorem 3. Define $\mathbf{Y} = \widehat{\mathbf{M}} = \mathbf{M} + \mathbf{E}$, where $\mathbf{E} = \widehat{\mathbf{M}} - \mathbf{M}$, and for $i \neq j = 1, 2$,

$$\mathbf{X}_1 = \mathbf{A}_i,\ \mathbf{X}_2 = (\mathbf{A}_j, \mathbf{B}).$$

We can then define

$$\mathbf{X}_{ij} = (\mathbf{X}_1, \mathbf{X}_2),\ \mathbf{D}_{ij} = \mathrm{diag}(\mathbf{D}_i, \mathbf{D}_j, \mathbf{0}),$$

where $\mathbf{D}_i$ contains r_i eigenvalues of the same order of magnitude specified in (6.23). Then $\mathbf{M} = \mathbf{X}_{ij}\mathbf{D}_{ij}\mathbf{X}_{ij}'$, and

$$\mathbf{X}_{ij}'\mathbf{Y}\mathbf{X}_{ij} = \begin{pmatrix} \mathbf{D}_i & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_j & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} + \begin{pmatrix} \mathbf{E}_{ii} & \mathbf{E}_{ij} & \mathbf{E}_{i3} \\ \mathbf{E}_{ij}' & \mathbf{E}_{jj} & \mathbf{E}_{j3} \\ \mathbf{E}_{i3}' & \mathbf{E}_{j3}' & \mathbf{E}_{33} \end{pmatrix},$$

where $\mathbf{E}_{ij} = \mathbf{A}_i'\mathbf{E}\mathbf{A}_j$ for $i, j = 1, 2, 3$, with $\mathbf{A}_3 = \mathbf{B}$. Hence

$$\mathbf{Y}_{11} = \mathbf{D}_i + \mathbf{E}_{ii}, \quad \mathbf{Y}_{12} = (\mathbf{E}_{ij}, \mathbf{E}_{i3}), \quad \mathbf{Y}_{22} = \begin{pmatrix} \mathbf{D}_j + \mathbf{E}_{jj} & \mathbf{E}_{j3} \\ \mathbf{E}_{j3}' & \mathbf{E}_{33} \end{pmatrix}.$$

We need to find the order of $\|\mathbf{E}_{ij}\|$ for $i, j = 1, 2, 3$. With $\widehat{\mathbf{M}} = \widehat{\mathbf{W}}_y\widehat{\mathbf{W}}_y'$, where $\widehat{\mathbf{W}}_y = (\widehat{\mathbf{\Sigma}}_y(1), \cdots, \widehat{\mathbf{\Sigma}}_y(k_0))$ and

$$\begin{aligned} \widehat{\mathbf{\Sigma}}_y(k) = {} & \mathbf{A}_1\widehat{\mathbf{\Sigma}}_1(k)\mathbf{A}_1' + \mathbf{A}_1\widehat{\mathbf{\Sigma}}_{12}(k)\mathbf{A}_2' + \mathbf{A}_1\widehat{\mathbf{\Sigma}}_{1\epsilon}(k) \\ & + \mathbf{A}_2\widehat{\mathbf{\Sigma}}_2(k)\mathbf{A}_2' + \mathbf{A}_2\widehat{\mathbf{\Sigma}}_{21}(k)\mathbf{A}_1' + \mathbf{A}_2\widehat{\mathbf{\Sigma}}_{2\epsilon}(k) \\ & + \widehat{\mathbf{\Sigma}}_{\epsilon 1}(k)\mathbf{A}_1' + \widehat{\mathbf{\Sigma}}_{\epsilon 2}(k)\mathbf{A}_2' + \widehat{\mathbf{\Sigma}}_{\epsilon}(k), \end{aligned} \tag{S.12}$$

after some tedious algebra (details omitted) we have by condition C5' that

$$\begin{aligned} \|\mathbf{E}_{11}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_1(k)\widehat{\mathbf{\Sigma}}_1(k)' - \mathbf{\Sigma}_1(k)\mathbf{\Sigma}_1(k)\|) = O_P(p^2 n^{-1/2}), \\ \|\mathbf{E}_{12}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_1(k)\widehat{\mathbf{\Sigma}}_{21}(k)' - \mathbf{\Sigma}_1(k)\mathbf{\Sigma}_{21}(k)\|) = O_P(p^{2-\delta_2/2} n^{-1/2}), \\ \|\mathbf{E}_{13}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_1(k)\widehat{\mathbf{\Sigma}}_{\epsilon 1}(k)'\mathbf{B}\|) = O_P(p^2 n^{-1/2}), \\ \|\mathbf{E}_{22}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_{21}(k)\widehat{\mathbf{\Sigma}}_{21}(k)' - \mathbf{\Sigma}_{21}(k)\mathbf{\Sigma}_{21}(k)\|) = O_P(p^{2-\delta_2} n^{-1/2}), \\ \|\mathbf{E}_{23}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_{21}(k)\widehat{\mathbf{\Sigma}}_{\epsilon 1}(k)'\mathbf{B}\|) = O_P(p^{2-\delta_2/2} n^{-1/2}), \\ \|\mathbf{E}_{33}\| &= O_P(\|\mathbf{B}'\widehat{\mathbf{\Sigma}}_{\epsilon 1}(k)\widehat{\mathbf{\Sigma}}_{\epsilon 1}(k)'\mathbf{B}\|) = O_P(p^2 n^{-1}). \end{aligned} \tag{S.13}$$

Consider $i = 1, j = 2$. From (6.23), we have $\lambda_{\min}(\mathbf{D}_1) \asymp p^2$, and hence $\|\mathbf{E}_{11}\| = o_P(\lambda_{\min}(\mathbf{D}_1))$ by (S.13). This implies that $\lambda_j(\mathbf{Y}_{11}) \asymp p^2$ for $j =$

$1, \cdots, r_1$. On the other hand, $\|\mathbf{Y}_{22}\| = o_P(p^2)$. This is because $\|\mathbf{D}_2\| \asymp p^{2-2c\delta_2} = o_P(p^2)$, and from (S.13) all of $\|\mathbf{E}_{22}\|, \|\mathbf{E}_{23}\|$ and $\|\mathbf{E}_{33}\|$ are $o_P(p^2)$ also. Hence,

$$\mathrm{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}) \asymp p^2.$$

Also from (S.13), $\|\mathbf{Y}_{12}\| = \|(\mathbf{E}_{12}, \mathbf{E}_{13})\| \leq \|\mathbf{E}_{12}\| + \|\mathbf{E}_{13}\| = O_P(p^2 n^{-1/2})$. Thus we have $\|\mathbf{Y}_{12}\| = o_P(\mathrm{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}))$ and hence (S.10) is satisfied. This means we can use (S.11) to conclude that

$$\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(p^2 n^{-1/2}/p^2) = O_P(n^{-1/2}).$$

Now consider $i = 2, j = 1$. We need a more precise order for $\|\mathbf{E}_{22}\|$. If $\|\mathbf{W}_{21}\| = o(p^{1-\delta_2})$ (i.e. $c = 1$ in the definition of Theorem 3), then

$$\begin{aligned}
\|\mathbf{E}_{22}\| &= O_P(\|\widehat{\mathbf{\Sigma}}_{21}(k)\widehat{\mathbf{\Sigma}}_{21}(k)' - \mathbf{\Sigma}_{21}(k)\mathbf{\Sigma}_{21}(k)'\|) \\
&= O_P(\|\widehat{\mathbf{\Sigma}}_{21}(k) - \mathbf{\Sigma}_{21}(k)\| \cdot \|\mathbf{\Sigma}_{21}(k)\|) \\
&= o_P(p^{1-\delta_2/2} n^{-1/2} \cdot p^{1-\delta_2}) \\
&= o_P(p^{2-3\delta_2/2} n^{-1/2}).
\end{aligned}$$

But $\lambda_{\min}(\mathbf{D}_2) \asymp p^{2-2\delta_2}$ by (6.23), we have $\|\mathbf{E}_{22}\| = o_P(\lambda_{\min}(\mathbf{D}_2))$ since we assumed $\kappa_n = p^{\delta_2/2} n^{-1/2} \to 0$. Hence $\lambda_j(\mathbf{Y}_{11}) \asymp p^{2-2\delta_2}$ for $j = 1, \cdots, r_2$.

On the other hand, if $\|\mathbf{W}_{21}\| \asymp p^{1-c\delta_2}$ for $1/2 \leq c < 1$ and $\|\mathbf{W}_1\mathbf{W}_{21}'\| \leq q\|\mathbf{W}_1\|_{\min}\|\mathbf{W}_{21}\|$, $0 \leq q < 1$, then

$$\|\mathbf{E}_{22}\| = O_P(p^{1-\delta_2/2} n^{-1/2} \cdot p^{1-c\delta_2}) = O_P(p^{2-(c+1/2)\delta_2} n^{-1/2}).$$

But $\lambda_{\min}(\mathbf{D}_2) \asymp p^{2-2c\delta_2}$ by (6.23), we have $\|\mathbf{E}_{22}\| = o_P(\lambda_{\min}(\mathbf{D}_2))$ since we assumed $p^{c\delta_2} n^{-1/2} \to 0$. Hence $\lambda_j(\mathbf{Y}_{11}) \asymp p^{2-2c\delta_2}$ for $j = 1, \cdots, r_2$.

Using the Wielandt's inequality in Lemma S.1 on $\mathbf{Y}_{22}$, we have for $j = 1, \cdots, p - r$,

$$0 \leq \lambda_j(\mathbf{E}_{33}) - \lambda_{r_1+j}(\mathbf{Y}_{22}) \leq \frac{\lambda_1(\mathbf{E}_{13}\mathbf{E}_{13}')}{\lambda_{r_1}(\mathbf{D}_1 + \mathbf{E}_{11}) - \lambda_j(\mathbf{E}_{33})}.$$

We have proved that $\lambda_{r_1}(\mathbf{D}_1 + \mathbf{E}_{11}) \asymp p^2$ when considering the case $i = 1, j = 2$. Also, (S.13) implies that $\lambda_1(\mathbf{E}_{13}\mathbf{E}_{13}') = O_P(p^4 n^{-1})$ and $\lambda_j(\mathbf{E}_{33}) = O_P(p^2 n^{-1}) = o_P(p^2)$. These imply that the right hand side of the inequalities above has order $O_P(p^4 n^{-1}/p^2) = O_P(p^2 n^{-1})$. Hence we can conclude from the same inequalities that

$$\lambda_{r_1+j}(\mathbf{Y}_{22}) = O_P(p^2 n^{-1}),\ j = 1, \cdots, p - r.$$

We also have $\lambda_j(\mathbf{Y}_{22}) \asymp p^2$ for $j = 1, \cdots, r_1$. To see this, we can decompose

$$\mathbf{Y}_{22} = \mathbf{S}_1 + \mathbf{S}_2, \quad \text{where} \;\; \mathbf{S}_1 = \begin{pmatrix} \mathbf{D}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \; \mathbf{S}_2 = \begin{pmatrix} \mathbf{E}_{11} & \mathbf{E}_{13} \\ \mathbf{E}_{13}' & \mathbf{E}_{33} \end{pmatrix}.$$

Then for $j = 1, \cdots, r_1$,

$$\lambda_j(\mathbf{Y}_{22}) = \sigma_j(\mathbf{Y}_{22}) \geq \sigma_j(\mathbf{S}_1) - \sigma_1(\mathbf{S}_2) \asymp p^2,$$

since by (6.23) we have $\sigma_j(\mathbf{S}_1) = \lambda_j(\mathbf{D}_1) \asymp p^2$, and $\sigma_1(\mathbf{S}_2) \leq \|\mathbf{E}_{11}\| + 2\|\mathbf{E}_{13}\| + \|\mathbf{E}_{33}\| = o_P(p^2)$.

The rates we found for $\lambda_j(\mathbf{Y}_{11})$ and $\lambda_j(\mathbf{Y}_{22})$ allow us to conclude that

$$\text{sep}(\mathbf{Y}_{11}, \mathbf{Y}_{22}) \asymp \begin{cases} p^{2-2\delta_2}, & \text{if } \|\mathbf{W}_{21}\| = o(p^{1-\delta_2}); \\ p^{2-2c\delta_2}, & \text{if } \|\mathbf{W}_{21}\| \asymp p^{1-c\delta_2}, \; 1/2 \leq c < 1 \text{ and} \\ & \quad \|\mathbf{W}_1\mathbf{W}_{21}'\| \leq q\|\mathbf{W}_1\|_{\min}\|\mathbf{W}_{21}\|, \; 0 \leq q < 1, \end{cases}$$

since we have assumed $p^{c\delta_2}n^{-1/2} \to 0$, implying that $p^2n^{-1} = o(p^{2-2c\delta_2})$.

Finally,

$$\|\mathbf{Y}_{12}\| \leq \|\mathbf{E}_{12}\| + \|\mathbf{E}_{23}\| = O_P(p^{2-\delta_2/2}n^{-1/2}) = o_P(p^{2-2c\delta_2}),$$

since in Theorem 3 we assumed $\nu_n = p^{2-\delta_2/2}n^{-1/2}/p^{2-2c\delta_2} \to 0$. Hence (S.10) is satisfied, and (S.11) implies that

$$\|\widehat{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(p^{2-\delta_2/2}n^{-1/2}/p^{2-2c\delta_2}) = O_P(\nu_n).$$

This completes the proof for the one-step estimation.

For the two-step estimation, write $\mathbf{M}^* = (\mathbf{A}_2 \; \mathbf{B}^*)\mathbf{D}^*(\mathbf{A}_2 \; \mathbf{B}^*)'$, where $\mathbf{B}^*$ is the orthogonal complement of $\mathbf{A}_2$, and $\mathbf{D}^*$ is diagonal with $\mathbf{D}^* = \text{diag}(\mathbf{D}_2^*, \mathbf{0})$. The matrix $\mathbf{D}_2^*$ contains λ_j^* for $j = 1, \cdots, r_2$, so that (6.24) implies $\text{sep}(\mathbf{D}_2^*, \mathbf{0}) \asymp p^{2-2\delta_2}$.

If we can show that $\|\mathbf{E}^*\| = \|\widehat{\mathbf{M}}^* - \mathbf{M}^*\| = O_P(p^{2-2\delta_2}\kappa_n)$, then $\|\mathbf{E}^*\| = o_P(\text{sep}(\mathbf{D}_2^*, \mathbf{0}))$, as $\kappa_n \to 0$. Hence we can use Lemma 3 of Lam *et al.* (2010) to conclude that

$$\|\widetilde{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(\|\mathbf{E}_{21}^*\|/\text{sep}(\mathbf{D}_2^*, \mathbf{0})).$$

Since $\|\mathbf{E}_{21}^*\| = O_P(\|\mathbf{E}^*\|) = O_P(p^{2-2\delta_2}\kappa_n)$, we then have

$$\|\widetilde{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(p^{2-2\delta_2}\kappa_n/p^{2-2\delta_2}) = O_P(\kappa_n),$$

which is what we want. Hence it remains to show $\|\mathbf{E}^*\| = O_P(p^{2-2\delta_2}\kappa_n)$.

First note that $\mathbf{\Sigma}_{y^*}(k) = \mathbf{A}_2\mathbf{\Sigma}_2(k)\mathbf{A}_2'$ so that $\|\mathbf{\Sigma}_{y^*}(k)\| = O(p^{1-\delta_2})$, and $\widehat{\mathbf{\Sigma}}_{y^*}(k) = \widehat{\mathbf{H}}_1\widehat{\mathbf{\Sigma}}_y(k)\widehat{\mathbf{H}}_1$, where $\widehat{\mathbf{H}}_1 = \mathbf{I} - \widehat{\mathbf{A}}_1\widehat{\mathbf{A}}_1'$. Noting also that $\|\widehat{\mathbf{H}}_1\mathbf{A}_1\| = O_P(\|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\|) = O_P(n^{-1/2})$, in view of (S.12), we have after some tedious algebra (details omitted) that

$$\begin{aligned}\|\mathbf{E}^*\| &= O_P(\|\widehat{\mathbf{\Sigma}}_{y^*}(k) - \mathbf{\Sigma}_{y^*}(k)\| \cdot \|\mathbf{\Sigma}_{y^*}(k)\|)\\ &= O_P(\|\widehat{\mathbf{H}}_1\widehat{\mathbf{\Sigma}}_{\epsilon 2}(k)\mathbf{A}_2'\widehat{\mathbf{H}}_1\| \cdot p^{1-\delta_2})\\ &= O_P(p^{1-\delta_2/2}n^{-1/2} \cdot p^{1-\delta_2})\\ &= O_P(p^{2-3\delta_2/2}n^{-1/2}) = O_P(p^{2-2\delta_2}\kappa_n),\end{aligned}$$

which completes the proof of the theorem. □

For Theorem 4, we only need to prove Lemma 1 in the Appendix, and the rest of the proof is already presented.

Proof of Lemma 1. Similar to (6.17), with conditions C3, C5' and C6', we can show that (details omitted) for model (4.13) and $k = 1, \cdots, k_0$,

$$\begin{aligned}\|\mathbf{\Sigma}_y(k)\| &= O(\|\mathbf{\Sigma}_1(k)\|) = O(p),\\ \|\widehat{\mathbf{\Sigma}}_y(k) - \mathbf{\Sigma}_y(k)\| &= O_P(pn^{-1/2} + \|\widehat{\mathbf{\Sigma}}_\epsilon(k)\|) = O_P(pn^{-1/2}),\end{aligned}$$

where the last line follows from $\|\widehat{\mathbf{\Sigma}}_\epsilon(k)\| = O_P(pn^{-1})$ when $n = O(p)$ and conditions C7 and C8 are satisfied (see proof of Theorem 2 for details). Hence by (6.17),

$$\begin{aligned}\|\widehat{\mathbf{M}} - \mathbf{M}\| &= O_P(\|\mathbf{\Sigma}_y(k)\| \cdot \|\widehat{\mathbf{\Sigma}}_y(k) - \mathbf{\Sigma}_y(k)\|) = O_P(p^2n^{-1/2}),\\ \|\mathbf{M}\| &= O(\|\mathbf{\Sigma}_y(k)\|^2) = O(p^2).\end{aligned} \tag{S.14}$$

By Theorem 3, using the notations in (6.19), we have

$$\|\widehat{\boldsymbol{\gamma}}_j - \mathbf{a}_j\| \le \begin{cases} \|\widehat{\mathbf{A}}_1 - \mathbf{A}_1\| = O_P(n^{-1/2}), & \text{for } j = 1, \cdots, r_1;\\ \|\widehat{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(\nu_n), & \text{for } j = r_1+1, \cdots, r;\\ \|\widehat{\mathbf{B}} - \mathbf{B}\| = O_P(\nu_n), & \text{for } j = r+1, \cdots, p,\end{cases} \tag{S.15}$$

where the rate for $\|\mathbf{B} - \mathbf{B}\|$ can be found similar to the proof of Theorem 3, and is thus omitted. With (S.14) and (S.15), the decomposition (6.19) is dominated by $\|I_3\|, \|I_4\|, \|I_5\| = O_P(p^2n^{-1/2})$, so that $|\widehat{\lambda}_j - \lambda_j| = O_P(p^2n^{-1/2})$. This proves (i).

For $j = r_1+1, \cdots, r$, the same decomposition (6.19) together with (S.14) and (S.15) gives dominating terms $\|I_2\| = O_P(p^2\nu_n^2)$ and $\|I_4\| = O_P(p^2n^{-1/2})$, so that $|\widehat{\lambda}_j - \lambda_j| = O_P(p^2(n^{-1/2} + \nu_n^2))$. This proves (ii).

For $j = r+1, \cdots, p$, the decomposition (6.20) together with (S.14) and (S.15) gives dominating term $\|K_3\| = O_P(\|\widehat{\mathbf{B}} - \mathbf{B}\| \cdot \|\mathbf{M}\|) = O_P(p^2\nu_n^2)$. Hence $\widehat{\lambda}_j = O_P(p^2\nu_n^2)$. This proves (iii).

When condition C9 holds, the proof of (iii)' is exactly the same as the proof for Theorem 2(ii)', with $\delta = 0$ and $\|\mathbf{P}\| = O_P(\nu_n)$, the same as the rate for $\|\widehat{\mathbf{A}} - \mathbf{A}\|$ in Theorem 3. Hence the rate of $\widehat{\lambda}_j$ for $j \geq r+1$ is $O_P(p^{3/2}\nu_n)$, which is (iii)'. This completes the proof for the original procedure.

For the two-step procedure, the decomposition (6.19) and (6.20) are both valid, with $\widehat{\mathbf{M}}$ and $\mathbf{M}$ replaced by $\widehat{\mathbf{M}}^*$ and $\mathbf{M}^*$ respectively, and $\widehat{\boldsymbol{\gamma}}_j$ replaced by $\widehat{\boldsymbol{\gamma}}_j^*$, the unit eigenvector of $\widehat{\mathbf{M}}^*$ for the j-th largest eigenvalue λ_j^* of $\mathbf{M}^*$. We have shown in the proof of Theorem 3 that

$$
\begin{aligned}
\|\mathbf{M}^*\| &= O(p^{2-2\delta_2}),\\
\|\widehat{\mathbf{M}}^* - \mathbf{M}^*\| &= O_P(\|\widehat{\boldsymbol{\Sigma}}_{y^*}(k) - \boldsymbol{\Sigma}_{y^*}(k)\| \cdot \|\boldsymbol{\Sigma}_{y^*}(k)\|)\\
&= O_P(p^{2-3\delta_2/2}n^{-1/2}) = O_P(p^{2-2\delta_2}\kappa_n).
\end{aligned} \tag{S.16}
$$

Let $\mathbf{a}_j^*$ be such that $\mathbf{A}_2 = (\mathbf{a}_1^*, \cdots, \mathbf{a}_{r_2}^*)$, $\mathbf{B}^* = (\mathbf{a}_{r_2+1}^*, \cdots, \mathbf{a}_p^*)$. Then we can show that

$$
\|\widehat{\boldsymbol{\gamma}}_j^* - \mathbf{a}_j^*\| \leq \begin{cases} \|\widetilde{\mathbf{A}}_2 - \mathbf{A}_2\| = O_P(\kappa_n), & \text{if } j = 1, \cdots, r_2;\\ \|\widetilde{\mathbf{B}}^* - \mathbf{B}^*\| = O_P(\kappa_n), & \text{if } j = r_2+1, \cdots, p. \end{cases} \tag{S.17}
$$

For $j = 1, \cdots, r_2$, with (S.16) and (S.17), the dominating terms for the decomposition (6.19) are $\|I_3\|, \|I_4\|, \|I_5\| = O_P(p^{2-2\delta_2}\kappa_n)$, so that $|\widehat{\lambda}_j^* - \lambda_j^*| = O_P(p^{2-2\delta_2}\kappa_n)$.

For $j = r_2+1, \cdots, p$, with (S.16) and (S.17), all terms of the decomposition (6.20) have the same rate $O_P(p^{2-2\delta_2}\kappa_n^2)$, so that $\widehat{\lambda}_j^* = O_P(p^{2-2\delta_2}\kappa_n^2)$.

Claim (vi) of the theorem is proved exactly the same as part (iii) of Theorem 2, and is omitted. □